

Week-1(Example)

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1.Create and select the database

The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
1 CREATE DATABASE student_db;
2 USE student_db;
3 CREATE TABLE student_marks (
4     roll_no INT PRIMARY KEY,
5     name VARCHAR(50),
6     subject VARCHAR(50),
7     marks DECIMAL(5,2)
8 );
```

The Output window displays the following results:

#	Time	Action	Message	Duration / Fetch
1	22:14:41	CREATE DATABASE student_db	1 row(s) affected	0.063 sec
2	22:14:41	USE student_db	0 row(s) affected	0.000 sec
3	22:14:41	CREATE TABLE student_marks (roll_no INT PRIMARY K...	0 row(s) affected	0.062 sec

2.Create the table

The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
1 CREATE DATABASE student_db;
2 USE student_db;
3 CREATE TABLE student_marks (
4     roll_no INT PRIMARY KEY,
5     name VARCHAR(50),
6     subject VARCHAR(50),
7     marks DECIMAL(5,2)
8 );
9 INSERT INTO student_marks (roll_no, name, subject, marks) VALUES
10 (1, 'Ravi', 'Math', 85.50),
11 (2, 'Sita', 'Math', 92.75),
12 (3, 'Anil', 'Math', 78.40),
13 (4, 'Priya', 'Math', 88.90),
14 (5, 'Vijay', 'Math', 80.25),
15 (6, 'subbusin', 'aws', 98.50),
16 (7, 'dwaraka', 'dbms', 95.75),
17 (8, 'ranjani', 'english', 97.40),
18 (9, 'kavithamam', 'aws1', 99.90),
19 (10, 'seetha', 'azure', 82.25);
```

The Output window displays the following results:

#	Time	Action	Message	Duration / Fetch
2	22:14:41	USE student_db	0 row(s) affected	0.0
3	22:14:41	CREATE TABLE student_marks (roll_no INT PRIMARY...	0 row(s) affected	0.0
4	22:17:43	CREATE DATABASE student_db	Error Code: 1007. Can't create database 'student_db'; data...	0.0
5	22:17:43	USE student_db	0 row(s) affected	0.0
6	22:17:43	CREATE TABLE student_marks (roll_no INT PRIMARY...	Error Code: 1050. Table 'student_marks' already exists	0.0
7	22:17:43	INSERT INTO student_marks (roll_no, name, subject, mar...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.0

3. Insert 10 records

The screenshot shows the SQL Developer interface with a query window titled 'Week_1_example'. The query contains 10 INSERT statements followed by a SELECT statement. The results grid shows the first 10 rows of the student_marks table. The output window shows the execution log with messages for each statement.

```
14 (5, 'Vijay', 'Math', 80.25),
15 (6, 'subbusir', 'aws', 98.50),
16 (7, 'dwaraka', 'dbms', 95.75),
17 (8, 'ranjani', 'english', 97.40),
18 (9, 'kavithamam', 'aws1', 99.90),
19 (10, 'seetha', 'azure', 82.25);
20 • SELECT * FROM student_marks;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

Output Window:

#	Time	Action	Message	Duration / Fetch
7	22:17:43	INSERT INTO student_marks (roll_no, name, subject, mar...	10 row(s) affected. Records: 10 Duplicates: 0 Warnings: 0	0.031 sec
8	22:18:40	CREATE DATABASE student_db	Error Code: 1007. Can't create database 'student_db'; dat...	0.016 sec
9	22:18:40	USE student_db	0 row(s) affected	0.000 sec
10	22:18:40	CREATE TABLE student_marks (roll_no INT PRIMAR...	Error Code: 1050. Table 'student_marks' already exists	0.000 sec
11	22:18:40	INSERT INTO student_marks (roll_no, name, subject, mar...	Error Code: 1062. Duplicate entry '1' for key 'student_mar...	0.000 sec
12	22:18:40	SELECT * FROM student_marks LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.00

4. Aggregate functions

The screenshot shows the SQL Developer interface with a query window titled 'Week_1_example'. The query contains a SELECT statement followed by five aggregate function queries. The results grid shows the first row of the student_marks table. The output window shows the execution log with messages for each statement.

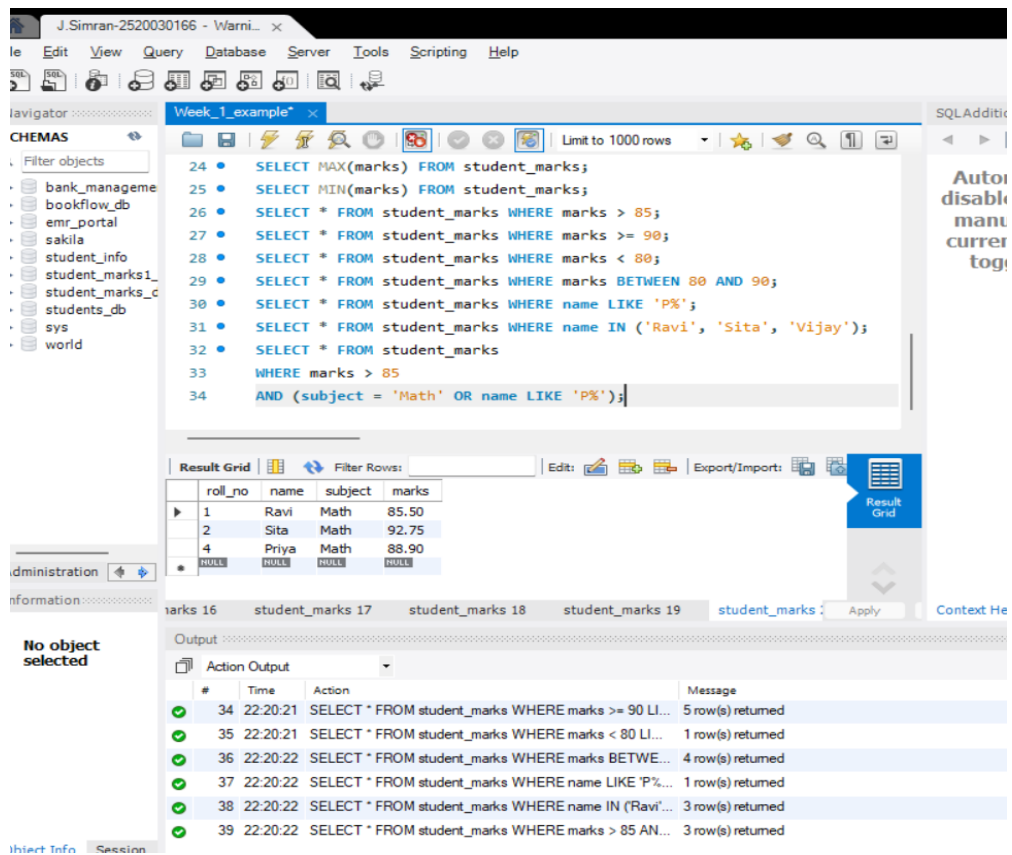
```
15 (6, 'subbusir', 'aws', 98.50),
16 (7, 'dwaraka', 'dbms', 95.75),
17 (8, 'ranjani', 'english', 97.40),
18 (9, 'kavithamam', 'aws1', 99.90),
19 (10, 'seetha', 'azure', 82.25);
20 • SELECT * FROM student_marks;
21 • SELECT COUNT(*) AS total_students FROM student_marks;
22 • SELECT SUM(marks) FROM student_marks;
23 • SELECT AVG(marks) FROM student_marks;
24 • SELECT MAX(marks) FROM student_marks;
25 • SELECT MIN(marks) FROM student_marks;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50

Output Window:

#	Time	Action	Message	Duration / Fetch
17	22:19:25	SELECT * FROM student_marks LIMIT 0, 1000	10 row(s) returned	0.00
18	22:19:25	SELECT COUNT(*) AS total_students FROM student_mar...	1 row(s) returned	0.00
19	22:19:25	SELECT SUM(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.00
20	22:19:25	SELECT AVG(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.00
21	22:19:25	SELECT MAX(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.00
22	22:19:26	SELECT MIN(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned	0.00

5. Filtering



SQL Developer interface showing a query window with 12 SQL statements for filtering data from the student_marks table. The results grid shows 4 rows of data. The output window shows the execution results for each statement.

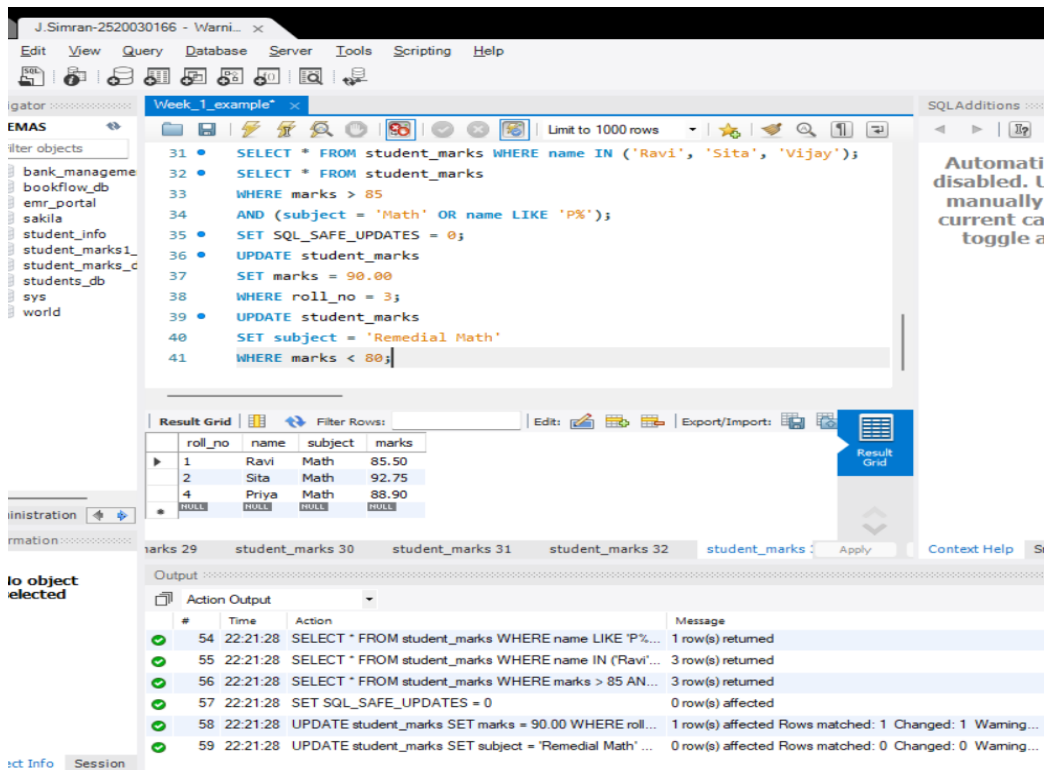
```
24 • SELECT MAX(marks) FROM student_marks;
25 • SELECT MIN(marks) FROM student_marks;
26 • SELECT * FROM student_marks WHERE marks > 85;
27 • SELECT * FROM student_marks WHERE marks >= 90;
28 • SELECT * FROM student_marks WHERE marks < 80;
29 • SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
30 • SELECT * FROM student_marks WHERE name LIKE 'P%';
31 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
32 • SELECT * FROM student_marks
33 WHERE marks > 85
34 AND (subject = 'Math' OR name LIKE 'P%');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
*	NULL	NULL	NULL

Output window showing execution results:

#	Time	Action	Message
34	22:20:21	SELECT * FROM student_marks WHERE marks >= 90 LI...	5 row(s) returned
35	22:20:21	SELECT * FROM student_marks WHERE marks < 80 LI...	1 row(s) returned
36	22:20:22	SELECT * FROM student_marks WHERE marks BETWE...	4 row(s) returned
37	22:20:22	SELECT * FROM student_marks WHERE name LIKE 'P%...	1 row(s) returned
38	22:20:22	SELECT * FROM student_marks WHERE name IN ('Ravi'...	3 row(s) returned
39	22:20:22	SELECT * FROM student_marks WHERE marks > 85 AN...	3 row(s) returned

6. Update records



SQL Developer interface showing a query window with 11 SQL statements for updating records in the student_marks table. The results grid shows 4 rows of data. The output window shows the execution results for each statement.

```
31 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
32 • SELECT * FROM student_marks
33 WHERE marks > 85
34 AND (subject = 'Math' OR name LIKE 'P%');
35 • SET SQL_SAFE_UPDATES = 0;
36 • UPDATE student_marks
37 SET marks = 90.00
38 WHERE roll_no = 3;
39 • UPDATE student_marks
40 SET subject = 'Remedial Math'
41 WHERE marks < 80;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90
*	NULL	NULL	NULL

Output window showing execution results:

#	Time	Action	Message
54	22:21:28	SELECT * FROM student_marks WHERE name LIKE 'P%...	1 row(s) returned
55	22:21:28	SELECT * FROM student_marks WHERE name IN ('Ravi'...	3 row(s) returned
56	22:21:28	SELECT * FROM student_marks WHERE marks > 85 AN...	3 row(s) returned
57	22:21:28	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
58	22:21:28	UPDATE student_marks SET marks = 90.00 WHERE roll...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
59	22:21:28	UPDATE student_marks SET subject = 'Remedial Math' ...	0 row(s) affected Rows matched: 0 Changed: 0 Warning...

7.Delete records

SQL script for deleting records:

```

33 WHERE marks > 85
34 AND (subject = 'Math' OR name LIKE 'P%');
35 SET SQL_SAFE_UPDATES = 0;
36 UPDATE student_marks
37 SET marks = 90.00
38 WHERE roll_no = 3;
39 UPDATE student_marks
40 SET subject = 'Remedial Math'
41 WHERE marks < 80;
42 DELETE FROM student_marks
43 WHERE roll_no = 5;
44 DELETE FROM student_marks

```

Result Grid:

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90

Action Output:

#	Time	Action	Message
77	22:22:41	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
78	22:22:41	UPDATE student_marks SET marks = 90.00 WHERE roll...	0 row(s) affected Rows matched: 1 Changed: 0 Warning...
79	22:22:41	UPDATE student_marks SET subject = 'Remedial Math' ...	0 row(s) affected Rows matched: 0 Changed: 0 Warning...
80	22:22:41	DELETE FROM student_marks WHERE roll_no = 5	1 row(s) affected
81	22:22:41	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected
82	22:22:41	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

8. Sorting

SQL script for sorting records:

```

41 WHERE marks < 80;
42 DELETE FROM student_marks
43 WHERE roll_no = 5;
44 DELETE FROM student_marks
45 WHERE marks < 75;
46 SET SQL_SAFE_UPDATES = 1;
47 SELECT * FROM student_marks
48 ORDER BY marks ASC;

```

Result Grid:

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90

Action Output:

#	Time	Action	Message
101	22:24:08	UPDATE student_marks SET marks = 90.00 WHERE roll...	0 row(s) affected Rows matched: 1 Changed: 0 Warning...
102	22:24:08	UPDATE student_marks SET subject = 'Remedial Math' ...	0 row(s) affected Rows matched: 0 Changed: 0 Warning...
103	22:24:08	DELETE FROM student_marks WHERE roll_no = 5	UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80
104	22:24:08	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected
105	22:24:08	SET SQL_SAFE_UPDATES = 1	0 row(s) affected
106	22:24:08	SELECT * FROM student_marks ORDER BY marks ASC ...	9 row(s) returned
107	22:24:08	SELECT * FROM student_marks ORDER BY marks DES...	9 row(s) returned
108	22:24:08	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
109	22:24:08	SELECT * FROM student_marks ORDER BY name DES...	9 row(s) returned

9.GROUP BY

The screenshot shows the SQL Developer interface with a query window titled "Week_1_example*". The query contains three SQL statements:

```
54 ORDER BY name DESC;SELECT subject, SUM(marks) AS total_marks
55 FROM student_marks
56 GROUP BY subject;
57 SELECT subject, AVG(marks) AS avg_marks
58 FROM student_marks
59 GROUP BY subject;
60 SELECT subject, COUNT(*) AS num_students
61 FROM student_marks
62 GROUP BY subject;
```

The "Result Grid" shows the output of the first query:

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1

The "Output" pane shows the execution log with the following messages:

#	Time	Action	Message
131	22:26:54	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected
132	22:26:54	SET SQL_SAFE_UPDATES = 1	0 row(s) affected
133	22:26:54	SELECT * FROM student_marks ORDER BY marks ASC ...	9 row(s) returned
134	22:26:54	SELECT * FROM student_marks ORDER BY marks DES...	9 row(s) returned
135	22:26:54	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
136	22:26:54	SELECT * FROM student_marks ORDER BY name DES...	9 row(s) returned
137	22:26:54	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned
138	22:26:54	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned
139	22:26:54	SELECT subject, COUNT(*) AS num_students FROM stu...	6 row(s) returned

10.HAVING

The screenshot shows the SQL Developer interface with a query window titled "Week_1_example*". The query contains three SQL statements:

```
62 GROUP BY subject;
63 SELECT subject, AVG(marks) AS avg_marks
64 FROM student_marks
65 GROUP BY subject
66 HAVING AVG(marks) > 90;
67 SELECT subject, COUNT(*) AS num_students
68 FROM student_marks
69 GROUP BY subject
70 HAVING COUNT(*) > 1;
```

The "Result Grid" shows the output of the first query:

subject	num_students
Math	4

The "Output" pane shows the execution log with the following messages:

#	Time	Action	Message
195	22:28:12	SELECT * FROM student_marks ORDER BY marks ASC ...	9 row(s) returned
196	22:28:12	SELECT * FROM student_marks ORDER BY marks DES...	9 row(s) returned
197	22:28:12	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
198	22:28:12	SELECT * FROM student_marks ORDER BY name DES...	9 row(s) returned
199	22:28:12	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned
200	22:28:12	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned
201	22:28:12	SELECT subject, COUNT(*) AS num_students FROM stu...	6 row(s) returned
202	22:28:12	SELECT subject, AVG(marks) AS avg_marks FROM stud...	4 row(s) returned
203	22:28:12	SELECT subject, COUNT(*) AS num_students FROM stu...	1 row(s) returned